RideLink

Smart Bus Tracking & Passenger Monitoring System

Electrical and Electronics Engineering

HNDE - Labuduwa

Presented by GROUP 03

RideLink

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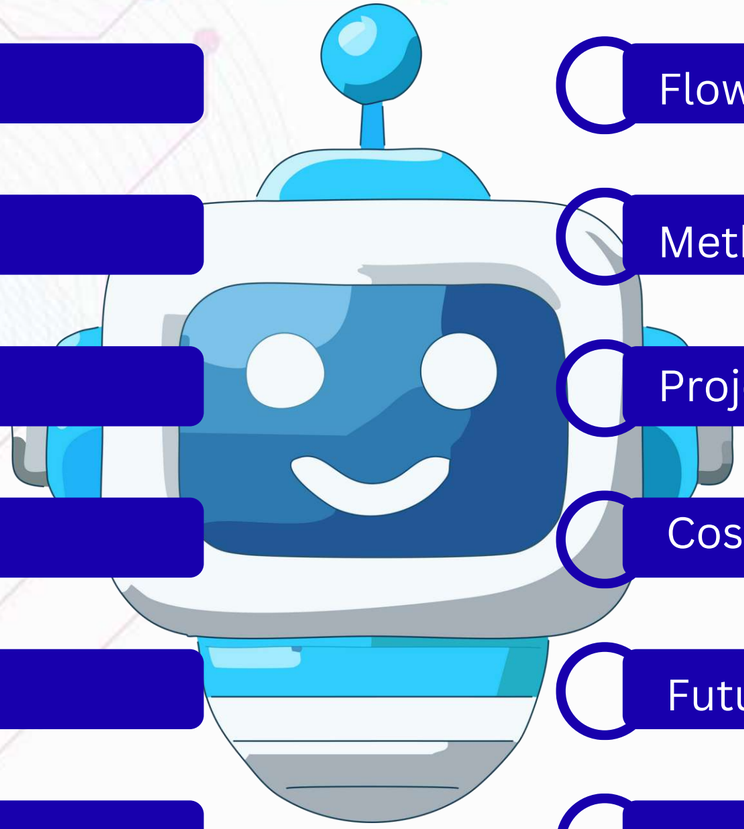
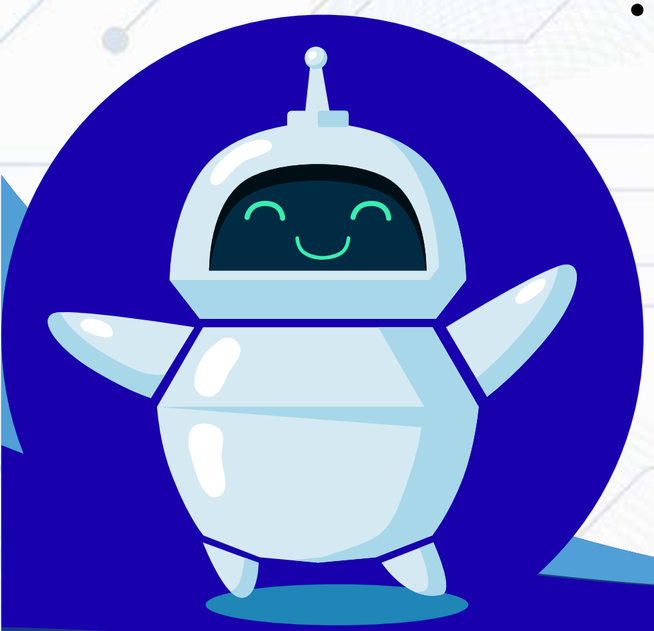


Introduction of Project

- Public transportation users often face uncertainty due to the lack of real-time bus tracking and passenger availability information. Passengers waste time at bus stops without knowing when the bus will arrive or whether it is already full.
- RideLink is an IoT-based smart transportation monitoring system that provides real-time bus tracking and automatic passenger counting using GPS, IR sensors, ESP32 microcontrollers and cloud technologies.



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Project Scope

- Real-time bus tracking
- Automatic passenger counting
- Web-based dashboard for passengers
- Admin and owner monitoring panels
- Cloud data storage
- Low-cost open-source implementation

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Project Aim

To design and implement an IoT-based smart bus monitoring system

To provide real-time location tracking and passenger occupancy information

To support efficient fleet management through cloud-based dashboards

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Objectives

- Track bus locations in real time
- Count passengers automatically
- Upload live data to cloud
- Display data through dashboards
- Provide admin and owner access
- Maintain scalability and low cost

Similar Systems

System	Description
Google Maps Transit	Location tracking only
RFID ticketing	No live tracking
Manual counting	Inaccurate
GPS vehicle trackers	No passenger monitoring

Components

ESP 32 Microcontroller

NEO 6M GPS Module

IR Sensor

OLED Display

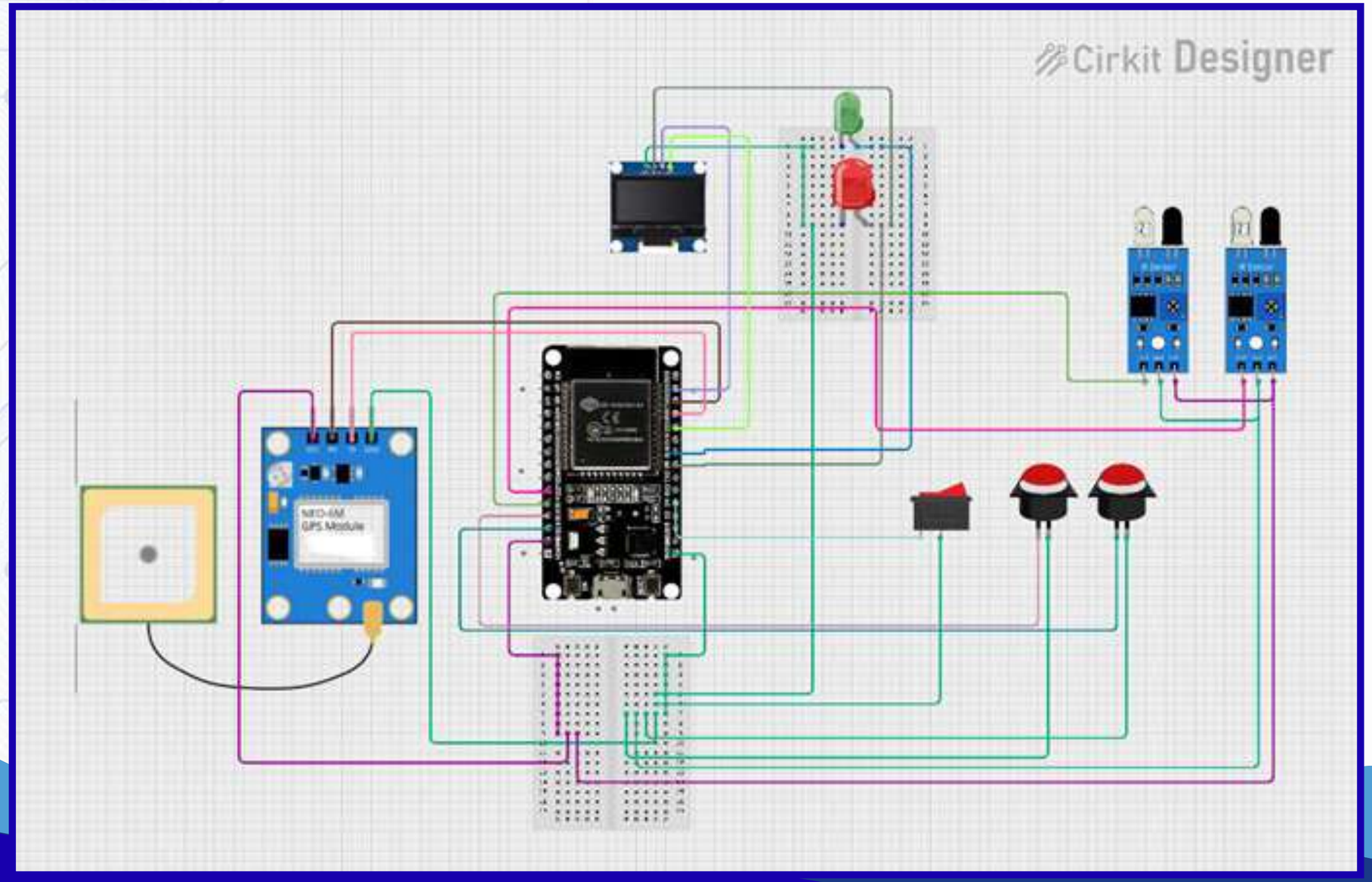
Firebase

Firebase Database

Frontend Dashboard

Node Express Backend Server

Circuit Diagram



Arduino Code

test3 | Arduino IDE 2.3.6

File Edit Sketch Tools Help

ESP32 Dev Module

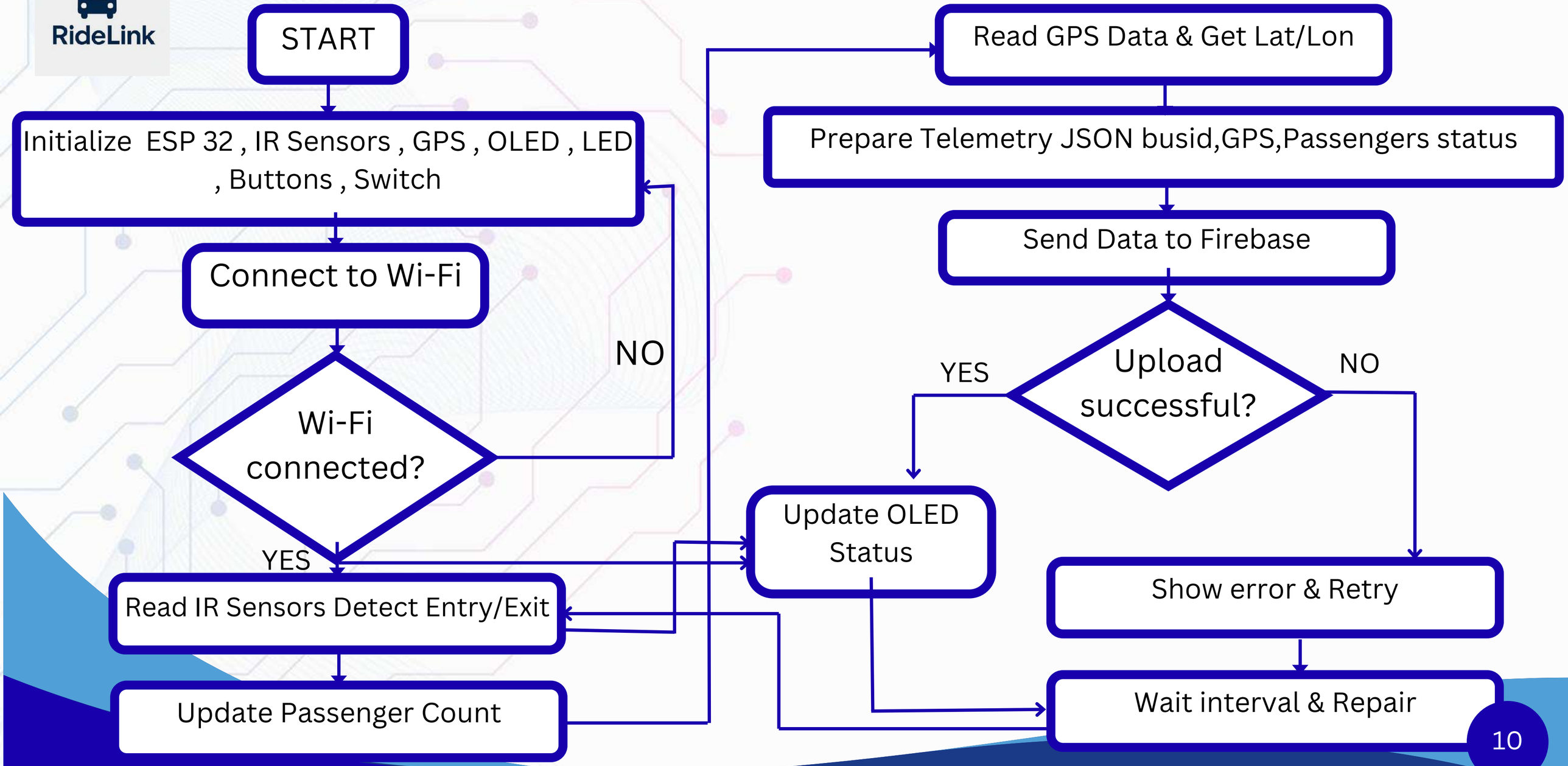
test3.ino

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include "time.h"
#include <sys/time.h>

const char* ssid      = "SLT-ADSL-8E582";
const char* password  = "ub2272098#";
const char* ntpServer = "pool.ntp.org";
const long  gmtOffset_sec = 19800;
const int   daylightOff
                        Loading...

const int SWITCH_PIN      = 4;   // Master switch
const int RED_LED_PIN     = 5;   // Online LED
const int GREEN_LED_PIN   = 18;  // WiFi LED
const int PUSHBTN_SEND    = 12;  // Force Send to Firebase
const int PUSHBTN_RESET   = 13;  // Reset Count

// --- SENSOR PINS ---
const int OUTER_PIR_PIN   = 14;  // OUTER Sensor
const int INNER_PIR_PIN   = 27;  // INNER Sensor

String busId = "RidelinkBus0001";
String ownerId = "RidelinkOwner005";
String firebaseRoot = "https://ridelink-26c32-default-rtdb.firebaseio.com/buses/" + busId;
String passengerApiUrl = "http://192.168.1.2:4002/api/update-bus";

// ========================
// OLED Setup
// ========================
```

Output

Ln 24, Col 1 ESP32 Dev Module on COM11 [not connected]

System Arrangement Flow Chart



Methodology

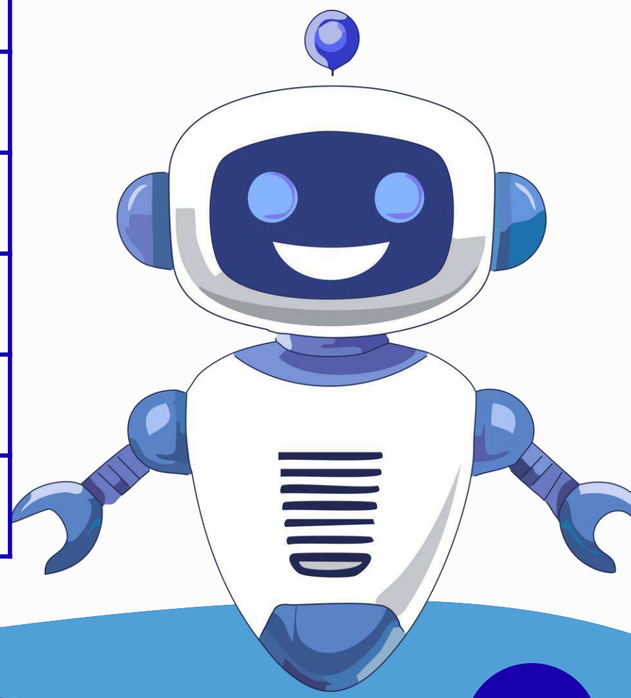
- Install ESP32, GPS and IR sensors inside buses
- Program firmware to collect passenger and location data
- Configure Firebase cloud database
- Develop backend server using Node.js
- Create web dashboard using React & Leaflet
- Test system accuracy and reliability

Project Time Plan

Task	September		October					November			December				Jan
Weeks	1	2	1	2	3	4	5	1	2	3	1	2	3	4	1
Selection Of Project & Preliminary Research															
Literature Review & Methodology															
Design & Planning															
Implementation															
Testing & Optimization															
Documentation															
Final Presentation Preparation															
Final Presentation & Submission															
Final Demonstration at Exhibition Level															

Cost Analyze

Components	Qty	Cost (Rs.)
ESP32 Dev Board	1	1400
NEO-6M GPS Module	1	1400
IR Break-Beam Sensors	2	360
OLED Display	1	850
Enclosure Box	1	420
Push Button	2	160
Others	-	1410
Total cost of the project		6000



Future Developments

- Develop a mobile application for passengers
- Implement AI-based arrival time prediction
- Introduce QR/RFID ticketing system
- Enable 4G/5G communication
- Add emergency alert system
- Implement solar-powered units

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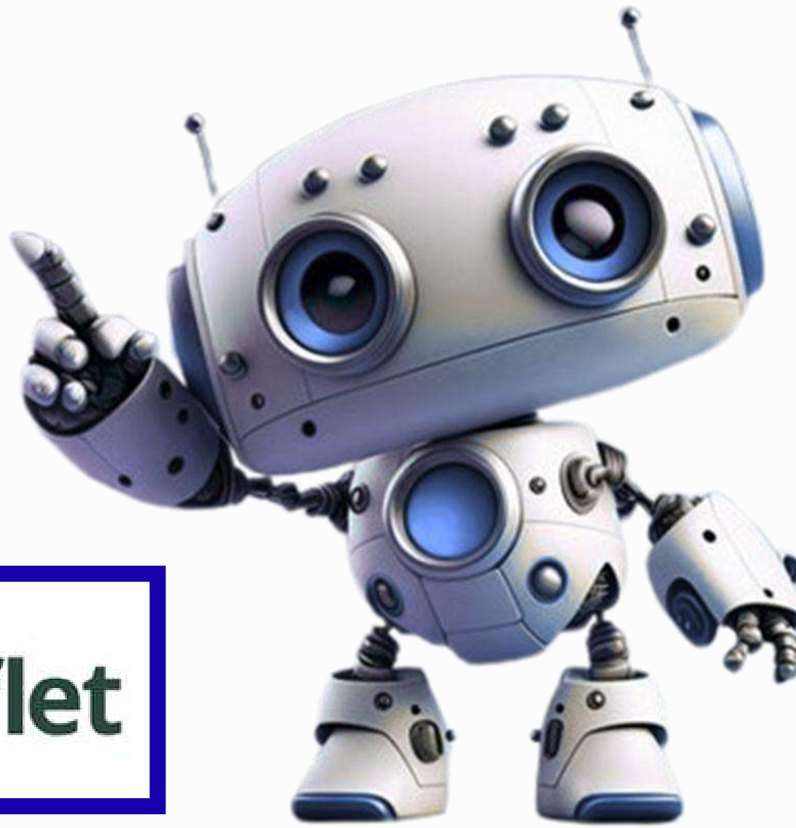
Resources

Arduino

Firebase

Cirkit Designer



Node Express

React Leaflet

References

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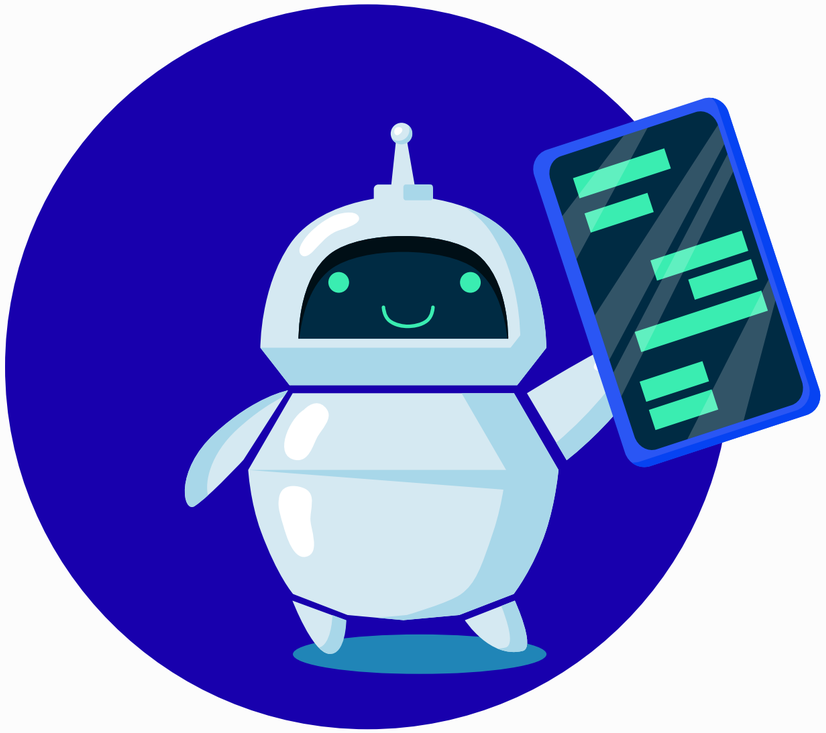
Q & A Session



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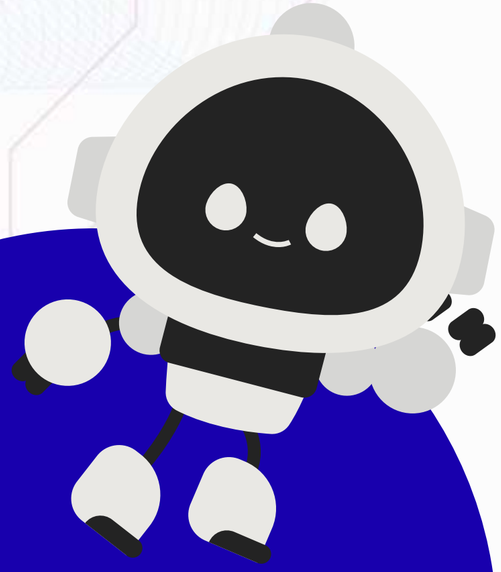
Thank you